HENGJI ZHANG

Quantitative Research · Investment Data Engineering · Machine Learning Systems | Market-data due diligence, point-in-time validation, factor verification, LLM benchmark design

bosprimigenious@bupt.edu.cn · github.com/bosprimigenious · bosprimigenious.github.io · Beijing, China

PROFILE

Computer-science undergraduate working at the intersection of market data and machine-learning systems. Two research/data internships in securities (alternative-data tracking at a sell-side institute; investment-data verification at an asset manager) plus a part-time benchmark-quality engagement, alongside four ship-and-ship engineering roles: a multimodal integrity platform (CLIP + Milvus), a Chinese AI-text detection platform built end to end, a university LLM practice platform, and a first-author detection framework under review at NeurIPS 2026. Habit of building the measurement before the claim: every number carries a source ID, a publication date and a reproducible workbook.

CURRENT ROLES

Independent Expert Engagement — Financial Benchmark Quality

Part-time Expert Reviewer (LLM evaluation) · Jun 2024 – Present · Remote

- Review 1,300+ model responses for market-data query benchmarks: rubric scoring, ground-truth cross-checks, and 180+ cross-reviewer adjudications; contributed sample screening, criterion decomposition, rubric calibration and dispute post-mortems.
- Enforce six mandatory checks: PIT availability by publication date; entity-mapping validity windows including cross-market duplicate codes; unit and frequency consistency (index points vs percent, TEU vs FEU, daily vs weekly); coverage numerator/denominator alignment; missing-value taxonomy instead of zero-fill; feasibility of documented replacement sources.
- Maintain a hallucination case library of 260+ instances (snapshot passed as PIT, missing values zero-filled, wrong coverage denominator, uncitable source, silent definition swap), classified by error type, trigger condition, decision boundary and correct handling; issued 110+ root-cause reports that fed back into the annotation standard.
- Compiled replacement-source registers for 70+ discontinued, low-frequency or broken series, each recording channel, update cadence, definitional gap versus the original indicator and a comparability verdict.

WorldQuant BRAIN — Alpha Research

Alpha research and factor validation (concurrent) · Jun 2023 – Present · Remote

- Construct cross-sectional and event-driven factors from price-volume, fundamental and alternative inputs (freight, utilisation, passenger flow, tenders); every factor carries a registry entry for input fields, frequency, unit, availability date and source permission.
- Evaluate single factors with IC / RankIC, layered backtests, long-short spread and turnover; retain decay curves across holding periods, and archive factor definition, universe and industry-neutralisation scheme together so results from different parameterisations are never pooled.
- Test robustness with out-of-sample windows, parameter grids and year-by-market-cap stability, guarding against reporting the best in-sample parameter set as the result.
- Convert turnover into commission and market-impact cost to report net-of-cost return and capacity, not gross spread.
- Control look-ahead and survivorship: align strictly to publication date, use point-in-time index constituents, and archive delisted or removed names separately.
- Gate factor admission on data acceptance — coverage, missing-value type and update frequency bound the usable sample and therefore IC reliability, so sources that fail acceptance never enter factor computation.

CITIC Securities · Asset Management

Investment Data Verification Intern · Mar 2023 – Present · Beijing

- Verify the factual queries behind investment-research output: 1,200+ alternative and market indicators (freight rates, capacity utilisation, production schedules, tender awards, satellite and passenger-flow series) checked under a four-step template — raise, reconcile, evidence, conclude.
- Run three-way source comparison across Wind, Bloomberg, iFinD and third-party alternative vendors; classify every discrepancy into five causes (definition, update timing, entity mapping, unit/precision, discontinuation) so a recurring break is fixed at the definition level instead of re-checked every cycle.
- Authored 80+ methodology documents stating definition, formula, frequency, unit, publication calendar, change log, known defects and applicability boundary; these became the acceptance basis for downstream queries.
- Maintain a change-and-sample-adjustment ledger (date, before/after value, affected scope) and a discontinuation register holding the last valid publication date, preventing stale snapshots from being consumed as current data.

EXPERIENCE, REVERSE-CHRONOLOGICAL

Information Network Engineering Research Centre, Ministry of Education

Engineering Intern — MingChaWanXiang research-integrity platform · Jan 2026 – Jun 2026 · Beijing

- Contributed to a multimodal research-integrity platform spanning text, figures, formulas, tables and citation networks, targeting paper-mill output, duplicate submission, image reuse and AIGC artefacts.
- Built multi-stage recall and similarity retrieval with CLIP embeddings stored in Milvus, supporting reuse localisation and side-by-side comparison rather than a single opaque score.
- Integrated LLM reasoning into the review flow and worked on risk visualisation, detection-report generation and evidence presentation; every exported finding is traceable to a source location.

TextGuard — Chinese AI-generated-text detection platform

Founder and Technical Lead (undergraduate innovation programme) · Nov 2025 – Jul 2026 · Beijing

- Designed and shipped a full-stack detection platform: Python / FastAPI inference service, React / TypeScript front end, risk visualisation, explanation panel, sample comparison and exportable detection reports; platform live.
- Defined the product around evidence rather than a verdict — each detection exposes the features that drove it, the comparison sample used, and the report that a reviewer can re-check.
- Led a small team through design, implementation, model training experiments, evaluation logging and project reporting; filed for the undergraduate innovation programme with paper, software-copyright and patent outputs planned.

Information Network Engineering Research Centre, Ministry of Education

Full-stack Intern — Zhilian 3.0, university LLM practice platform · Jul 2025 – Jan 2026 · Beijing

- Owned one vertical slice from page to API in Vue 3 + TypeScript: list, filter, pagination and record-version views, with multi-condition state carried in the query string.
- Aligned field definitions, error codes and record versions with the backend, and surfaced empty-result, no-permission and timeout states explicitly so a cold start is never mistaken for an outage.
- Worked to a "reproducible from the public site" bar — institute review score 90.00.

Shenwan Hongyuan Securities · Research Institute

Cyclical Industries Research Intern (transport and commodities) · Feb 2021 – Feb 2023 · Beijing

- Co-maintained high-frequency alternative databases — container freight (SCFI / CCFI / BFI families), blast-furnace and cement utilisation, order books and production schedules, tender award values, freight volume and passenger flow — supporting 240+ weekly notes and data commentaries across 10+ sub-sectors.
- Built the indicator ledger: publishing entity, frequency, history start, release time and licensing constraint per series; it later became the denominator source for coverage checks.
- Cross-validated 1,500+ alternative indicators, reconciling same-source definition variants (composite vs route-level, tax-inclusive vs exclusive, MoM vs YoY base) and flagging definition changes and sample-universe shifts.
- Applied seasonal adjustment and base restoration to freight-rate and utilisation series, and separated initial prints, revised prints and forecast vintages in working papers so a revised value was never presented as a first release.

RESEARCH AND PUBLICATION

TextGuard: A Dual-Path Semantic–Statistical Robust Framework for Chinese AI-Generated Text Detection

Hengji Zhang et al. (student first author / co-first author) · NeurIPS 2026 — under review

- Problem. Chinese AI-generated-text detectors generalise poorly out of distribution. Concatenating statistical features with a semantic encoder creates a shortcut: the statistical head carries most of the in-distribution signal, then collapses when the generator or domain shifts, because the two representations are fused without any gating on the semantic path.
- Method. Two components replace naive concatenation. CDFI (Cross-Domain Feature Injection) treats five LPP statistics as a query over the semantic representation and injects them through residual gating instead of channel concatenation, so statistical evidence modulates the encoder rather than bypassing it. GA-AT (Granularity-Aware Adversarial Training) applies PGD perturbations at character, word and sentence granularity, forcing the decision boundary to stay stable under local rewrites that humans consider meaning-preserving.
- Features. A parallel Erlangshen + CNN–BiLSTM branch computes five interpretable LPP signals — four-character collocation, structural-particle usage, punctuation entropy, type-token ratio and sentence-length distribution — chosen for their known asymmetry between human and generated Chinese text.
- Leakage control. Training uses HC3 only; no benchmark sample is ever seen during fitting, and the evaluation suites are held out at document level rather than chunk level.
- Evaluation. TextGuard-Bench defines five evaluation axes; MGAB supplies a held-out attack set; G3 quantifies the gap between in-distribution and out-of-distribution AUC, so a model that wins in-distribution but degrades under shift is not rewarded. Baselines: BERT, RADAR and a Chinese paraphrase-uncertainty detector.
- Results. Holdout n = 33,739: accuracy 98.75%, AUC 0.9992, MCC 0.9746. Question-answer subset n = 3,516: AUC 0.950.
- My contribution. LPP feature design, the MGAB held-out protocol, and the cross-dataset OOD evaluation design; co-authored the method and wrote the evaluation section.

EDUCATION

Beijing University of Posts and Telecommunications (BUPT)

B.Eng. in Computer Science and Technology · Sep 2024 – Jun 2028 (expected) · Beijing

- Rank 47 / 452 (top 10%) in the major.
- Coursework: Data Structures and Course Design 100, Object-Oriented Programming 100, Mathematical Modelling 99, Introduction to Computers and Programming 97, Big Data Practice 95, Intelligent Robotics Practice 92.
- Campus roles: psychological-support committee member, Class 301, School of Computer Science (2024 – present; campus top-10 psychological-support volunteer, nominee, 2026); university middle/long-distance running team (2024 – 2025); 123 hours of volunteering.
- Interests: middle/long-distance running (5 km PB 18:20), marathons, football, tennis, reading, writing fiction; self-taught French.

HONOURS AND AWARDS

- National-level competition awards: 2 · Provincial / ministerial: 8 · University-level: 4
- University second-class scholarship; Merit Student (■■■■) — 1 each

TECHNICAL SKILLS

Languages	Python (primary), TypeScript / JavaScript, SQL, Java (coursework), Bash
ML / data	PyTorch, pandas / NumPy, scikit-learn, CLIP, Milvus vector search, LLM inference and prompt evaluation
Web / services	FastAPI, React, Vue 3, Vite, REST / JSON, auth and error-code design, pagination and versioning
Finance data	Wind, Bloomberg, iFinD, WorldQuant BRAIN; PIT alignment, coverage and denominator design, entity-mapping validity, missing-value taxonomy, factor validation and cost-adjusted returns

CERTIFICATIONS AND TESTS

CFA: Level 2

Languages (tests): IELTS 6.5; CET-4 and CET-6 passed